

09: Inserting Data into the Database

You can insert data into a database by executing SQL INSERT statements within your Java code.

- For non-select operations like INSERT, UPDATE, and DELETE, the `executeUpdate()` method is used.
- The `executeUpdate()` method returns an integer value indicating the number of rows affected by the SQL operation. If the value is zero, it means no rows were affected, implying the data insertion was unsuccessful.

👉 Steps for Inserting the Data:

1. Create a SQL INSERT statement:

```
String sql = "INSERT INTO studentInfo(id, sname, sage, scity) VALUES (1, 'Rohan', 17, 'Bengaluru')";
```

Here, studentInfo is the table with id, name, sage, scity as columns.

2. Execute the SQL statement:

- Use the Statement object's `executeUpdate()` method to execute the INSERT statement.
- The `executeUpdate()` method returns an integer representing the number of rows affected by the SQL operation.

```
int rowAffected = statement.executeUpdate(sql);
```

3. Check If Insertion Was Successful:

After executing the INSERT statement, check the value of the rowAffected variable to determine whether the insertion was successful.

```
if (rowAffected > 0) {  
    System.out.println("Insertion successful! Number of rows affected: " +  
        rowAffected);  
} else {  
    System.out.println("Insertion failed. No rows were affected.");  
}
```

Code:

```
import java.sql.*;

public class LaunchApp1 {

    public static void main(String[] args) {

        String url = "jdbc:mysql://localhost:3306/yourDatabaseName";

        String username = "yourUsername";

        String password = "yourPassword";


        String sql = "INSERT INTO studentInfo(id, sname, sage, scity) VALUES
(1, 'Rohan', 17, 'Bengaluru')";


        Connection conn = null;
        Statement stmt = null;

        try {

            conn = DriverManager.getConnection(url, username, password);

            stmt = conn.createStatement();

            int rowAffected = stmt.executeUpdate(sql);

            if (rowAffected > 0) {

                System.out.println("Insertion successful! Number of rows affected: "
+ rowAffected);

            } else {

                System.out.println("Insertion failed. No rows were affected.");

            }

        } catch (SQLException e) {

            e.printStackTrace();

        } finally {
```

```
try {  
    if (stmt != null) stmt.close();  
    if (conn != null) conn.close();  
} catch (SQLException e) {  
    e.printStackTrace();  
}  
}  
}
```